Dependent Position Determining – General Data

1. General

The dependent position determining aids can be comprised of Automatic Direction Finding System (ADF), Distance Measuring Equipment (DME), VHF Omni-Range (VOR) and transponder systems, dependent on customer requirements.

A. VHF Navigation

Refer to 34-51-00, Multi-Mode Digital Radio (MMDR) – General Data and 34-52-00, VHF Navigation System – General data for system details.

The VHF navigation systems 1 and 2 receive VHF Omni-Range (VOR), Localizer (LOC) and Glide Slope (GS) signals. The systems process:

- The VOR signals to give course navigation data to a VOR ground station
- The LOC and GS signals to show deviation along the flightpath of the instrument landing systems.

The VHF navigation systems 1 and 2 have these components:

- VHF/NAV 1/GS 1 receivers (integrated in the No. 1 MMDR system)
- VHF/NAV 2/GS 2 receivers (integrated in the No. 2 MMDR system)
- A NAV antenna with antenna coupler (splitter)
- A Glideslope antenna

B. Distance Measuring Equipment (DME)

Refer to 34-54-00, Distance Measuring Equipment (DME) (Mod 6/2049) – General Data for system details.

The Distance Measuring Equipment (DME) gives the distance from the aircraft to a DME ground station. The dual DME consists of two transceivers (DME 1 and DME 2) and one antenna per DME transceiver. DME 1 antenna and DME 2 antenna. Provision has been made for the dual DMEs to be installed on the aircraft, this is part of the IFR Operations Bundle customer Order Option, S.O.O. 6209.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 there is one DME transceiver (DME 1) and one antenna installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the second DME transceiver (DME 2) and associated antenna are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6386.

C. Automatic Direction Finding System (ADF)

Refer to 34-55-00, Automatic Direction Finding System (ADF) – General Data for system details.

The Automatic Direction Finder (ADF) is used to display the aircrafts relative bearing to a tuned ground station.

The ADF tunes frequencies from 190 kHz to 1,799 kHz and 2,180 kHz to 2,189 kHz. Functions consist of ADF, ANT, and BFO tuning modes. ADF 1 and optional ADF 2 antennas include both a loop and sense component and are installed at FS297.7 on the top of the fuselage.

The ADF system is an integral component of the KTR 2280 Multi-Mode Digital Radio (MMDR) (refer to 34-51-00, Multi-Mode Digital Radio – General Data).

For aircraft with S.O.O. 6388 incorporated, the Automatic Direction Finding (ADF) system is not installed.

For aircraft with S.O.O. 6388 incorporated, the single Automatic Direction Finding (ADF) and associated antenna are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6405 and dual

Automatic Direction Finding (ADF) and associated antennas are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6217.

D. KXP 2290 Transponder System

Refer to 34-53-00, KXP 2290 Transponder System – General Data for system details.

The Mode S Transponder (KXP 2290) (Pre Mod 6/2303) replies to Air Traffic Control Radar Beacon System (ATCRBS) Mode A and C, Intermode, and Mode S interrogations. Provisions have been made for the KXP 2290 Transponder System, this is part of the IFR operational bundle customer order option, S.O.O. 6209.

Aircraft incorporating Mod 6/2354 upgrade the transponders from KXP 2290 to KXP 2290A, with optional Automatic Dependent Surveillance – Broadcast (ADS-B) Out capability. The KXP 2290A transponders are direct replacements to the KXP 2290 LRU's, installed behind each of the pilot's PFD Controller.

The ADS-B out functionality is enabled through the incorporation of customer Standard Order Option, S.O.O. 6347. SBAS GPS Receivers (Mod 6/2304) is a pre-requisite for S.O.O. 6347.

Each transponder system includes the KXP 2290 or KXP 2290A LRU located behind the associated PFD Controller and bottom and top (diversity) antennas.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 there is one KXP 2290A Transponder installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the second KXP 2290A Transponder and associated antennas are installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6381.

E. Global Positioning System (GPS)

Refer to 34-56-00, Global Positioning System (GPS) – General Data for system details.

Two independent KGS 200 Global Navigation System Sensor Unit (GNSSU) receivers are installed in the nose avionics bay. The receiver use 24 channels, grouped in 3 hardware groups, each consisting of 8 channels. One group of 8 channels is used for navigation at any given time. The GPS sensor unit calculates and outputs navigation data, satellite measurement data, Receiver Autonomous Integrity Monitoring (RAIM) and Predictive RAIM (PRAIM). The processed output data is sent to the CSIO module in the MAU, via ARINC 429, for use by the rest of the avionic system. The sensor data unit also manages Sign Status Matrix (SSM), satellite status and does Built-In Test Equipment (BITE) functions. An Apex maintenance function interfaces with the GPS sensor. The GPS data page can be accessed from the Sensors page. The Sensors page can be accessed with the systems MFD lower left window in focus and selecting the Sensors page menu. The TAWS unit receives GPS position via the MAU for accurate position determination.

Aircraft incorporating Mod 6/2304 have upgraded the KGS 200 Global Navigation System Sensor Unit (GNSSU) receivers for Satellite Based Augmentation System (SBAS) capable GPS receivers.

Provisions have been made for SBAS GPS approach capability to allow SBAS-based GPS approaches, such as LPV approaches, through the incorporation of customer Standard Order Option, S.O.O. 6225.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 there is one upgraded KGS 200 Global Navigation System Sensor Unit (GNSSU) receiver installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the second upgraded KGS 200 Global Navigation System Sensor Unit (GNSSU) receiver and associated antenna is installed on the aircraft by incorporating customer Standard Order Option S.O.O. 6380.

F. XM Satellite Weather System

Refer to 34-57-00, XM Satellite Weather System – General data for system details.

The XM Satellite Weather System is a streaming weather data receiver.

The XM Weather receiver is mounted on the rear avionics rack. The XM Satellite Weather system sends received weather data to the Modular Avionics Unit (MAU) Advanced Graphics Unit (AGU), which is then displayed on the Multi-Function Display (MFD).

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